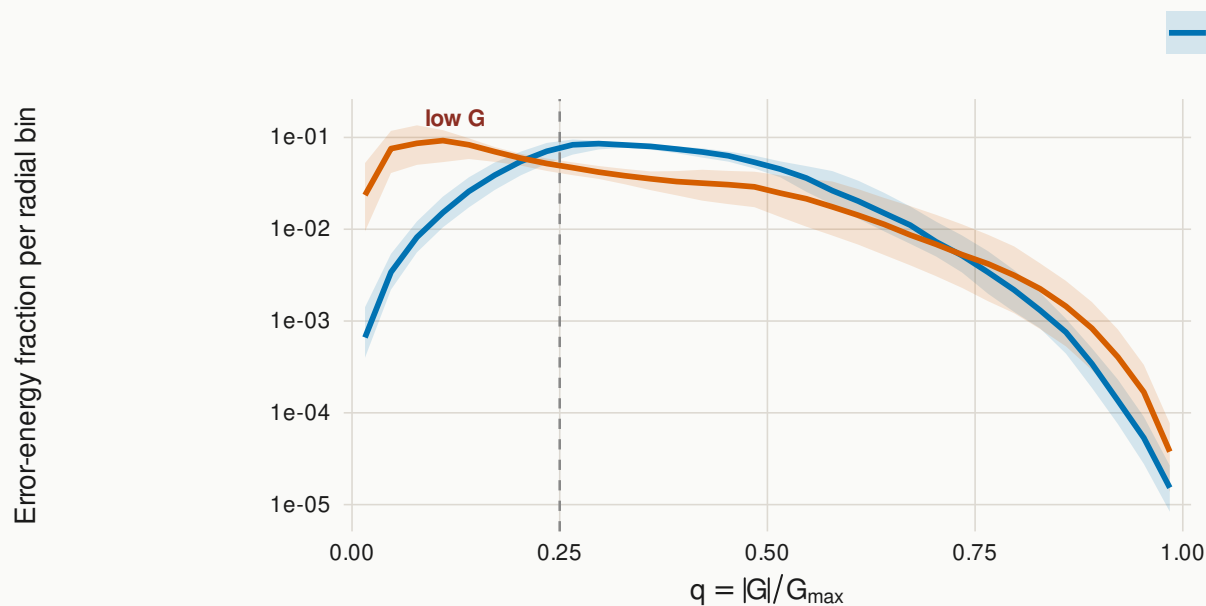


Figure 7 | Frequency allocation of reconstruction error controls Hartree fidelity at matched L_∞

Exact ZFP/SZ3 common-support audit: 457 matched pairs across 214 materials. The Nyquist-safe operator reproduces the historical codec effect while restoring exact Fourier/real-space parity.

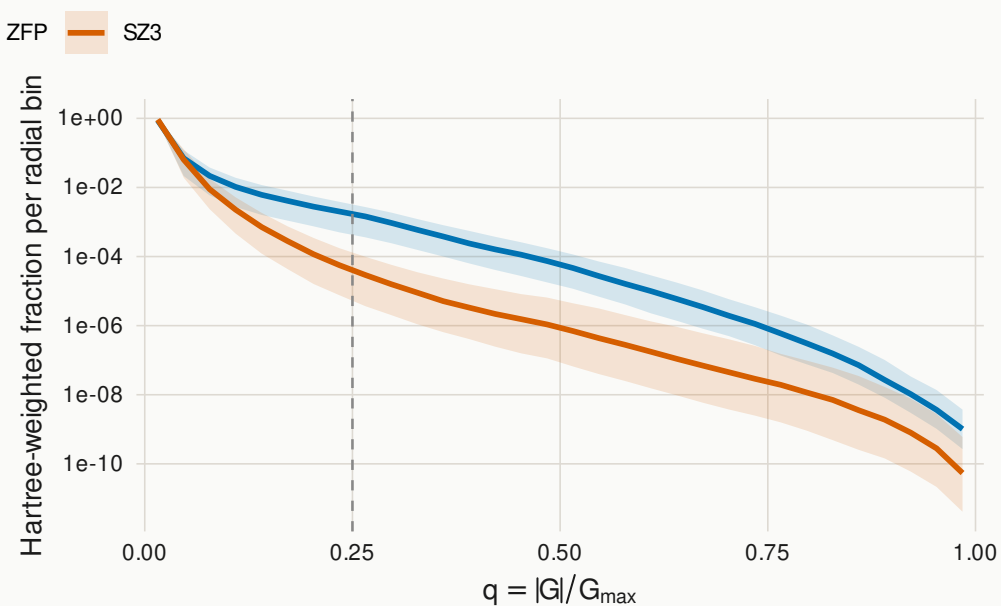
A | ZFP shifts reconstruction error away from low G

Median radial error-energy fraction; ribbons show pairwise IQR across 457 matched pairs



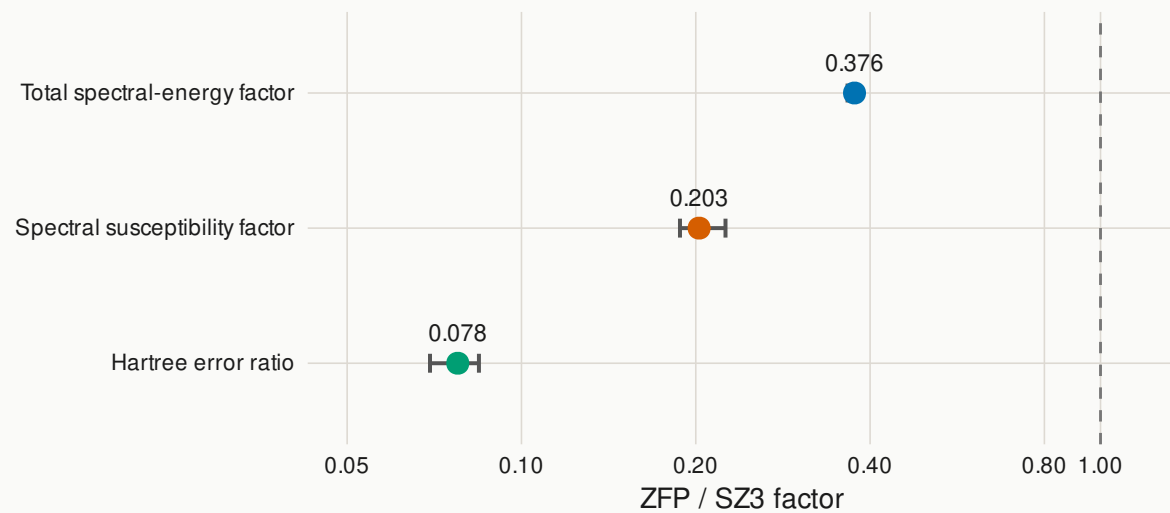
B | $|G|^{-4}$ weighting makes low-G error decisive

Median fraction of Hartree-weighted spectral energy in each radial bin



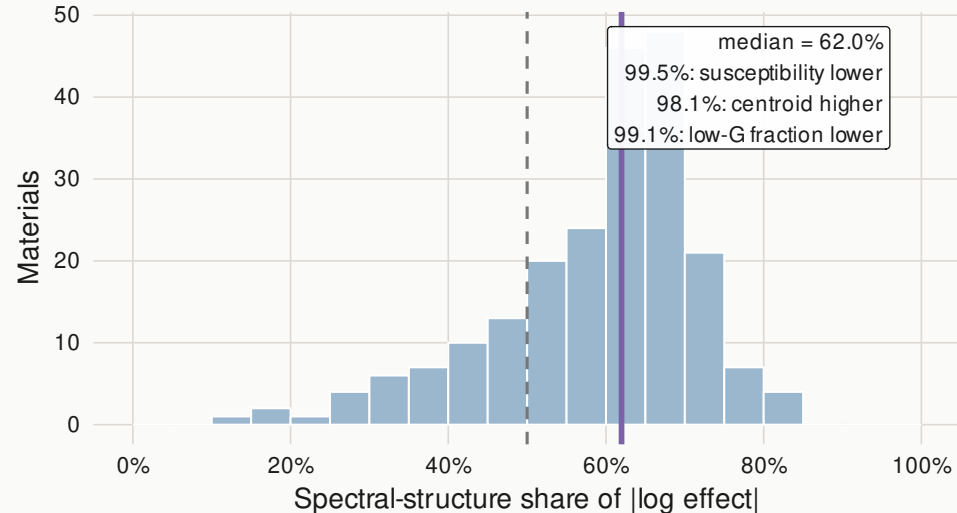
C | Decomposition of the Hartree codec effect

Material-level centers with 95% bootstrap intervals; exact closure is pairwise



D | Frequency structure dominates

Material-level share of the absolute-log effect from spectral susceptibility



Historical Hartree ratio = 0.0776221; Nyquist-safe ratio = 0.0776219; maximum Parseval relative error = 1.30×10^{-15} .
Exact pairwise identity: $R_H = \sqrt{E_{\text{ZFP}}/E_{\text{SZ3}}} \times \sqrt{S_H/\text{ZFP}/S_H/\text{SZ3}}$. Panel C reports separately aggregated material-level centers.